

Melanoma Skin Cancer Detection using Convolutional Neural Network

Tanha Kabir Koly

1 Introduction

Melanoma is one of the most aggressive and life-threatening forms of skin cancer, responsible for a large proportion of skin cancer-related deaths. Early detection significantly improves survival rates; however, manual diagnosis using dermoscopic images is time-consuming and subject to human error.

Recent advancements in deep learning have enabled automated medical image analysis systems that can assist clinicians in early diagnosis. However, several challenges remain, including limited dataset size, class imbalance, and the need to prioritize sensitivity (recall) over overall accuracy.

This study aims to develop and evaluate deep learning models for melanoma detection using dermoscopic images. The primary contributions of this work include:

- A comparative analysis of CNN-based and transfer learning-based approaches
- Investigation of class imbalance handling techniques such as focal loss and oversampling
- Exploration of threshold optimization for improving classification performance

2 Literature Review

Traditional image classification approaches relied on handcrafted features such as texture, color, and shape descriptors. These approaches often lacked robustness and generalization capabilities.

The introduction of Convolutional Neural Networks (CNNs) enabled automatic feature extraction, leading to significant improvements in performance. However, training CNNs from scratch requires large datasets, which are often unavailable in medical imaging.

Transfer learning has emerged as an effective solution, where pretrained models such as EfficientNet leverage knowledge from large-scale datasets (e.g., ImageNet). EfficientNet, in particular, introduces a compound scaling method that balances network depth, width, and resolution, achieving superior performance with fewer parameters.

Another major challenge in medical datasets is class imbalance. Since melanoma cases are relatively rare, models tend to be biased toward the majority class. Techniques such as focal loss and oversampling have been proposed to address this issue by emphasizing minority class samples.

3 Methodology

The overall methodology adopted in this study is illustrated in Figure 1. A progressive experimental framework was designed to evaluate different deep learning strategies, starting from a basic CNN model and advancing toward more sophisticated approaches incorporating transfer learning and class imbalance handling techniques.

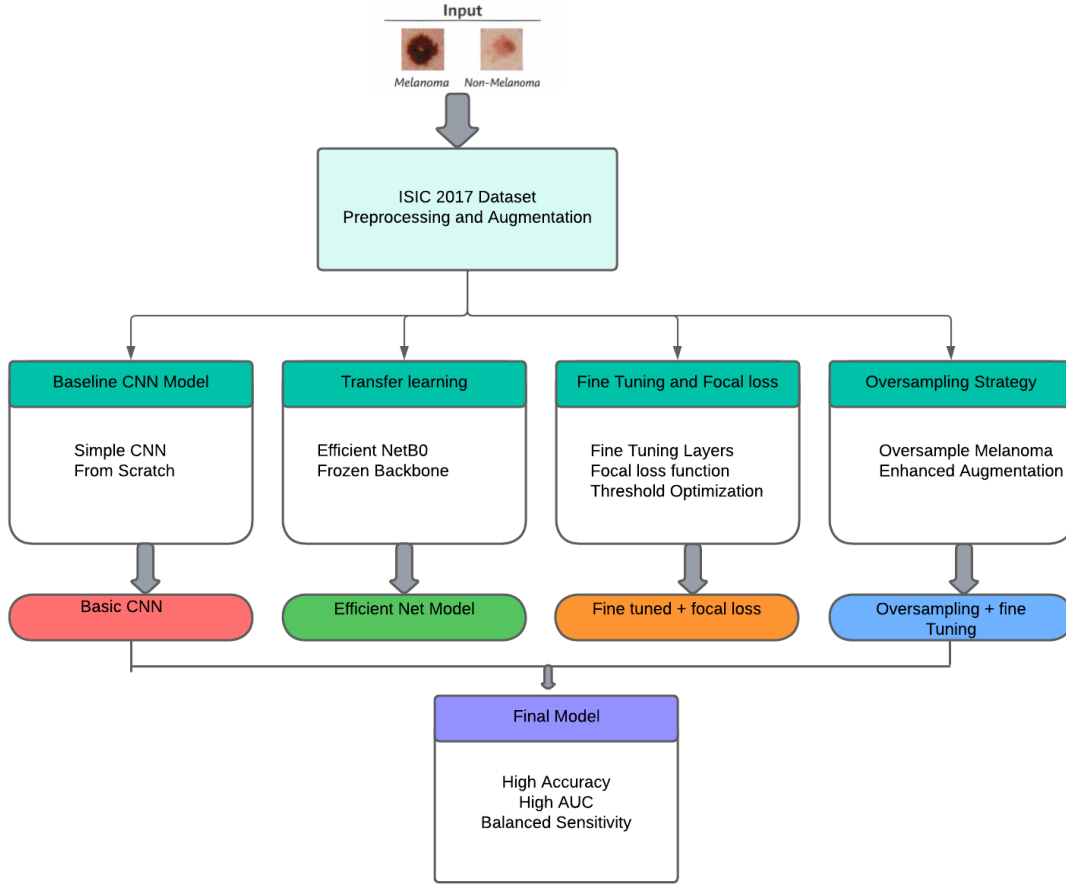


Figure 1: Proposed methodology for melanoma detection. The pipeline illustrates the progression from a baseline CNN model to transfer learning with EfficientNet, followed by fine-tuning with focal loss and threshold optimization, and finally oversampling-based training to achieve balanced performance.

3.1 Baseline CNN Model

A basic CNN architecture was implemented as the initial baseline. The model consists of:

- Three convolutional layers with ReLU activation
- Max pooling layers for spatial downsampling
- A fully connected dense layer with dropout

This model is trained from scratch and serves as a reference for evaluating the benefits of transfer learning.

3.2 Transfer Learning using EfficientNetB0

To improve feature extraction, EfficientNetB0 pretrained on ImageNet was used. The convolutional backbone was initially frozen, and a custom classification head was added. This approach leverages learned representations from large datasets, improving generalization.

3.3 EfficientNet Architecture

EfficientNet is a family of convolutional neural network architectures designed to achieve high performance with optimal computational efficiency. Unlike traditional CNN scaling methods that arbitrarily increase network depth or width, EfficientNet introduces a *compound scaling* approach, which uniformly scales network depth, width, and input resolution using a fixed set of scaling coefficients.

The EfficientNetB0 model, used in this study, serves as the baseline architecture in the EfficientNet family. It is pretrained on the ImageNet dataset and consists of a sequence of convolutional layers combined with Mobile Inverted Bottleneck Convolution (MBConv) blocks. These blocks are derived from MobileNet architectures and are designed to reduce computational complexity while maintaining strong feature representation capability.

Each MBConv block consists of:

- An expansion phase, where the number of channels is increased
- A depthwise convolution, which applies spatial filtering separately to each channel
- A squeeze-and-excitation (SE) module, which adaptively recalibrates channel-wise feature responses
- A projection phase, which reduces the dimensionality back to a lower channel size

The use of depthwise separable convolutions significantly reduces the number of parameters and computational cost compared to standard convolutions. Additionally, the squeeze-and-excitation mechanism enhances the network’s ability to focus on the most informative features.

EfficientNetB0 follows a structured architecture composed of multiple stages, each containing several MBConv blocks with increasing depth and feature channels. As the network progresses, spatial resolution decreases while feature richness increases, allowing the model to capture both low-level and high-level features.

In this study, the EfficientNetB0 model was used as a pretrained feature extractor. Initially, the convolutional backbone was frozen to retain the learned ImageNet features. A custom classification head was then added, consisting of:

- Global Average Pooling layer
- Fully connected dense layer with ReLU activation
- Dropout layers for regularization
- Final sigmoid output layer for binary classification

Subsequently, fine-tuning was performed by unfreezing the top layers of the EfficientNet backbone. This allowed the model to adapt high-level features specifically to dermoscopic image characteristics, improving classification performance.

The use of EfficientNetB0 provides several advantages:

- Strong feature extraction capability due to pretrained weights
- Efficient parameter utilization through compound scaling
- Improved generalization on small datasets

These properties make EfficientNet particularly suitable for medical image classification tasks, where data is often limited and high accuracy is required.

3.4 Fine-Tuning and Focal Loss

To further enhance performance:

- The top layers of EfficientNet were unfrozen for fine-tuning
- Focal loss was used to address class imbalance

The focal loss function is defined as:

$$FL(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t) \quad (1)$$

where α balances class importance and γ focuses on hard-to-classify samples.

3.5 Threshold Optimization

Instead of using a fixed classification threshold (0.5), threshold tuning was performed based on F1-score to achieve an optimal balance between precision and recall.

3.6 Oversampling Strategy

In the final model, the minority class (melanoma) was oversampled to create a balanced training dataset. This approach reduces bias toward the majority class and improves overall classification performance.

4 Implementation

4.1 Dataset

The ISIC 2017 dataset was used, containing dermoscopic images labeled as melanoma or non-melanoma.

Initially, the Complete MedNode dataset was considered, which contains approximately 100 melanoma and 70 non-melanoma images. However, due to its limited size, it was excluded from the final experiments. Training deep learning models on such small datasets increases the risk of overfitting and limits generalization capability.

4.2 Preprocessing and Augmentation

The following preprocessing steps were applied:

- Image resizing to 224–300 pixels
- Normalization using EfficientNet preprocessing
- Data augmentation:
 - Random horizontal and vertical flips
 - Brightness and contrast adjustment
 - Saturation variation

4.3 Training Strategy

Training was performed in two stages:

- Stage 1: Frozen backbone training
- Stage 2: Fine-tuning of top layers

Additional techniques include:

- Early stopping to prevent overfitting
- Learning rate scheduling
- Batch size of 16–32

4.4 Evaluation Metrics

The models were evaluated using:

- Accuracy
- ROC-AUC
- Precision, Recall, F1-score
- Confusion Matrix

5 Experimental Results

5.1 Performance Comparison

Model	Accuracy	AUC	Recall (Melanoma)	Precision (Melanoma)
Basic CNN	Low	Low	Low	Low
EfficientNet	71.33%	0.74	53.33%	35.56%
Fine-tuned + Focal Loss	71.33%	0.75	86.67%	40.00%
Oversampling + Fine-tuning	77.33%	0.786	60.00%	45.00%

5.2 Analysis of Results

The baseline CNN model performed poorly due to limited data and lack of pretrained features. Transfer learning significantly improved performance by leveraging learned representations.

The focal loss model achieved the highest recall for melanoma detection (86.67%), indicating its effectiveness in identifying minority class samples. However, it increased false positives.

The oversampling-based model achieved the best overall performance, balancing precision and recall while improving accuracy and AUC.

5.3 Trade-off Discussion

The results highlight an important trade-off:

- High recall models reduce missed melanoma cases
- Balanced models improve overall classification accuracy

In medical applications, minimizing false negatives is often more critical than maximizing accuracy.

6 Discussion

This study demonstrates that model performance is highly dependent on how class imbalance is handled. Simply improving accuracy is insufficient for medical diagnosis tasks.

The use of focal loss and oversampling provides different advantages:

- Focal loss emphasizes hard examples
- Oversampling improves dataset balance

Although the Complete MedNode dataset was initially considered, its small size limited its usefulness for deep learning. Future work could explore combining multiple datasets to improve robustness.

7 Conclusion

A comprehensive evaluation of deep learning models for melanoma detection was conducted. The final model achieved 77.33% accuracy and 0.786 ROC-AUC, while another configuration achieved significantly higher melanoma recall (86.67%).

These findings highlight the importance of selecting models based on application-specific priorities, particularly in medical contexts where sensitivity is critical.

8 Future Work

Future improvements may include:

- Using larger datasets such as ISIC 2018/2020
- Combining datasets for better generalization
- Incorporating explainability techniques (e.g., Grad-CAM)
- Exploring advanced architectures such as Vision Transformers

9 Code Availability and Reproducibility

All implementation details, including source code, readme file, are made publicly available to support the reproducibility of this study. The repository can be accessed at:

https://drive.google.com/drive/folders/1avL-2maApVd3BmGaI7zIrCGPphUhagUU?usp=drive_link

References

- [1] L. R. Soenksen, T. Kassis, S. T. Conover, B. Marti-Fuster, J. S. Birkenfeld, J. Tucker-Schwartz, A. Naseem, R. R. Stavert, C. C. Kim, M. M. Senna, *et al.*, “Using deep learning for dermatologist-level detection of suspicious pigmented skin lesions from wide-field images,” *Science Translational Medicine*, vol. 13, no. 581, p. eabb3652, 2021.
- [2] S. A. Leachman, P. B. Cassidy, S. C. Chen, C. Curiel, A. Geller, D. Gareau, G. Pellacani, J. M. Grichnik, J. Malvey, J. North, *et al.*, “Methods of melanoma detection,” *Melanoma*, pp. 51–105, 2016.
- [3] A. Astorino, A. Fuduli, P. Veltri, and E. Vocaturo, “Melanoma detection by means of multiple instance learning,” *Interdisciplinary Sciences: Computational Life Sciences*, vol. 12, no. 1, pp. 24–31, 2020.
- [4] R. L. Siegel, K. D. Miller, and A. Jemal, “Cancer statistics, 2018,” *CA: a cancer journal for clinicians*, vol. 68, no. 1, pp. 7–30, 2018.
- [5] M. E. Celebi, Q. Wen, H. Iyatomi, K. Shimizu, H. Zhou, and G. Schaefer, “A state-of-the-art survey on lesion border detection in dermoscopy images,” *Dermoscopy image analysis*, vol. 10, no. 1, pp. 97–129, 2015.
- [6] G. P. Guy Jr, S. R. Machlin, D. U. Ekwueme, and K. R. Yabroff, “Prevalence and costs of skin cancer treatment in the us, 2002- 2006 and 2007- 2011,” *American journal of preventive medicine*, vol. 48, no. 2, pp. 183–187, 2015.
- [7] A. G. Goodson and D. Grossman, “Strategies for early melanoma detection: Approaches to the patient with nevi,” *Journal of the American Academy of Dermatology*, vol. 60, no. 5, pp. 719–735, 2009.
- [8] K. Jayapriya and I. J. Jacob, “Hybrid fully convolutional networks-based skin lesion segmentation and melanoma detection using deep feature,” *International Journal of Imaging Systems and Technology*, vol. 30, no. 2, pp. 348–357, 2020.
- [9] M. A. Kassem, K. M. Hosny, and M. M. Fouad, “Skin lesions classification into eight classes for isic 2019 using deep convolutional neural network and transfer learning,” *IEEE Access*, vol. 8, pp. 114822–114832, 2020.
- [10] A. A. Ali and H. Al-Marzouqi, “Melanoma detection using regular convolutional neural networks,” in *2017 International Conference on Electrical and Computing Technologies and Applications (ICECTA)*, pp. 1–5, IEEE, 2017.
- [11] N. H. Quang *et al.*, “Automatic skin lesion analysis towards melanoma detection,” in *2017 21st Asia Pacific symposium on intelligent and evolutionary systems (IES)*, pp. 106–111, IEEE, 2017.
- [12] J. Daghrir, L. Tlig, M. Bouchouicha, and M. Sayadi, “Melanoma skin cancer detection using deep learning and classical machine learning techniques: A hybrid approach,” in *2020 5th international conference on advanced technologies for signal and image processing (ATSIP)*, pp. 1–5, IEEE, 2020.

- [13] R. Zhang, “Melanoma detection using convolutional neural network,” in *2021 IEEE International conference on consumer electronics and computer engineering (ICCECE)*, pp. 75–78, IEEE, 2021.
- [14] R. Ashraf, S. Afzal, A. U. Rehman, S. Gul, J. Baber, M. Bakhtyar, I. Mehmood, O.-Y. Song, and M. Maqsood, “Region-of-interest based transfer learning assisted framework for skin cancer detection,” *IEEE Access*, vol. 8, pp. 147858–147871, 2020.
- [15] F. Nunnari, C. Bhuvaneshwara, A. O. Ezema, and D. Sonntag, “A study on the fusion of pixels and patient metadata in cnn-based classification of skin lesion images,” in *International Cross-Domain Conference for Machine Learning and Knowledge Extraction*, pp. 191–208, Springer, 2020.
- [16] R. Kaur, H. GholamHosseini, R. Sinha, and M. Lindén, “Melanoma classification using a novel deep convolutional neural network with dermoscopic images,” *Sensors*, vol. 22, no. 3, p. 1134, 2022.
- [17] N. Gouda and J. Amudha, “Skin cancer classification using resnet,” in *2020 IEEE 5th International Conference on Computing Communication and Automation (ICCCA)*, pp. 536–541, IEEE, 2020.
- [18] B. Ozdemir and I. Pacal, “An innovative deep learning framework for skin cancer detection employing convnextv2 and focal self-attention mechanisms,” *Results in Engineering*, vol. 25, p. 103692, 2025.
- [19] A. T. P. Nguyen, R. M. Jewel, and A. Akter, “Comparative analysis of machine learning models for automated skin cancer detection: Advancements in diagnostic accuracy and ai integration,” *The American Journal of Medical Sciences and Pharmaceutical Research*, vol. 7, no. 01, pp. 15–26, 2025.
- [20] S. Afrifa, V. Varadarajan, P. Appiahene, T. Zhang, D. Gyamfi, and R.-M. O. M. Gyening, “Deep neural networks for skin cancer classification: Analysis of melanoma cancer data,” *Journal of Advances in Information Technology*, vol. 16, no. 1, 2025.